1 Solution — GIAM 1.4 Problem 14

1.1 Problem

Calculate the following quantities: (a) $3 \bmod 5$; (b) $37 \bmod 7$; (c) $1000001 \bmod 100000$; (d) $6 \operatorname{div} 6$; (e) $7 \operatorname{div} 6$; (f) $1000001 \operatorname{div} 2$.

1.2 Method

From Problem 13, every division writes $n = q\,d + r$ with $0 \le r < d$, where

$$n \operatorname{div} d \;=\; q \;=\; \left\lfloor \tfrac{n}{d} \right\rfloor \qquad \text{(the quotient)}, \qquad n \bmod d \;=\; r \qquad \text{(the remainder)}.$$

So each answer is found by writing $n = qd + r$ and reading off the requested part — r for "mod," q for "div."

1.3 The Six Computations

Expression	Write as $n = q\,d + r$	Read off	Answer
(a) $3 \bmod 5$	$3 = 0 \cdot 5 + \mathbf{3}$	r	3
(b) $37 \bmod 7$	$37 = 5 \cdot 7 + \mathbf{2}$	r	2
(c) $1000001 \bmod 100000$	$1000001 = 10 \cdot 100000 + \mathbf{1}$	r	1
(d) $6 \operatorname{div} 6$	$6 = \mathbf{1} \cdot 6 + 0$	q	1
(e) $7 \operatorname{div} 6$	$7 = \mathbf{1} \cdot 6 + 1$	q	1
(f) $1000001 \operatorname{div} 2$	$1000001 = \mathbf{500000} \cdot 2 + 1$	q	500000

Table 1.1

1.4 At a Glance

Write n = q·d + r (0 ≤ r < d). Then "mod" reads off r (orange); "div" reads off q (blue).

expression	n = q · d + r	answer
(a) 3 mod 5	3 = 0·5 + $\boxed{3}$	3
(b) 37 mod 7	37 = 5·7 + $\boxed{2}$	2
(*c*) 1000001 mod 100000	1000001 = 10·100000 + $\boxed{1}$	1
(d) 6 div 6	6 = $\boxed{1}$ ·6 + 0	1
(e) 7 div 6	7 = $\boxed{1}$ ·6 + 1	1
(f) 1000001 div 2	1000001 = $\boxed{500000}$ ·2 + 1	500000

mod → the remainder r   div → the quotient q

Figure 1.1: Six rows: for each, n is written as q·d + r. The 'mod' answers (a, b, c) highlight the remainder r in orange; the 'div' answers (d, e, f) highlight the quotient q in blue. Results: 3, 2, 1, 1, 1, 500000.

1.5 Remark — "mod" and "div" Are Two Halves of One Division

Each single division produces *both* a quotient and a remainder; "div" and "mod" just pick which one you keep. The two are linked by $n = (n \operatorname{div} d) \cdot d + (n \bmod d)$, so knowing one and the dividend gives the other. For instance, part (e) computed $7 \operatorname{div} 6 = 1$; the companion is $7 \bmod 6 = 7 - 1 \cdot 6 = 1$. And in (f), alongside $1000001 \operatorname{div} 2 = 500000$ sits $1000001 \bmod 2 = 1$ — the remainder of 1 being exactly the statement that 1000001 is **odd**.

1.6 Remark — Small Cases: When $n < d$

Part (a) illustrates the boundary case $n < d$: here $3 < 5$, so no copies of 5 fit, the quotient is 0, and the *whole* number is the remainder. In general,

$$n < d \implies n \operatorname{div} d = 0 \quad \text{and} \quad n \bmod d = n.$$

This is why $3 \bmod 5 = 3$ — the input passes through unchanged. (Contrast (d), the other boundary: when $d \mid n$ exactly, the remainder is 0 and div gives the clean quotient, here $6 \operatorname{div} 6 = 1$.)

1.7 Remark — Mod and Div by Powers of Ten Read and Drop Digits

Parts (c) and (f) reward a slick shortcut. Because $100000 = 10^5$, taking $\operatorname{mod} 10^5$ simply **reads off the last five decimal digits**, and $\operatorname{div} 10^5$ **drops** them:

$$1000001 \;=\; \underbrace{10}_{\text{div } 10^5}\;\underbrace{00001}_{\text{mod } 10^5}, \qquad 1000001 \bmod 100000 = \underbrace{00001} = 1.$$

(So $1000001 \operatorname{div} 100000 = 10$, the digits *above* the cut, and $1000001 \bmod 100000 = 1$, the digits *below*.) This is the decimal cousin of the binary trailing-bit facts from Problem 2: dividing by b^k shifts the base-b representation right by k places (keeping the high part), while mod b^k keeps the low k places. Recognising 100000 as 10^5 turns (c) into a one-glance answer — just look at the bottom five digits of 1000001, which are 00001.

1.8 Remark — A Sanity Check via the Defining Identity

Each answer can be re-verified instantly by plugging back into $n = qd + r$ and confirming $0 \le r < d$:

Part	**Check** $n = qd + r$	$0 \le r < d$?
(a)	$0 \cdot 5 + 3 = 3$ ✓	$0 \le 3 < 5$ ✓
(b)	$5 \cdot 7 + 2 = 37$ ✓	$0 \le 2 < 7$ ✓
(c)	$10 \cdot 100000 + 1 = 1000001$ ✓	$0 \le 1 < 100000$ ✓
(d)	$1 \cdot 6 + 0 = 6$ ✓	$0 \le 0 < 6$ ✓
(e)	$1 \cdot 6 + 1 = 7$ ✓	$0 \le 1 < 6$ ✓
(f)	$500000 \cdot 2 + 1 = 1000001$ ✓	$0 \le 1 < 2$ ✓

Table 1.2

Every row reconstructs the original n and keeps the remainder in range, so all six answers are confirmed.